

ECO387C - Fall 2024

Problem Set 7

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

Question 1 - McCall Model: comparative statics

Consider an unemployed worker who is infinitely lived and maximizes their expected discounted sum of utility:

$$\mathbb{E}_t \sum_{t=0}^{\infty} \beta^t c_t$$

with discount factor $\beta \in (0, 1)$.

Each period, the unemployed worker receives a wage offer w and can either:

1. Accept the offer and receive wage w in perpetuity, or
2. Reject the offer, receive unemployment benefit given by b , and receive a new wage draw for next period.

Wages are iid with probability mass function $p_1, \dots, p_N$, where p_i is the probability of observing wage offer w_i .

Assume that $w \sim U_{[10, 60]}$, $b = 20$, $\beta = 0.96$, and $N = 101$.

1. Calculate the reservation wage and plot the worker's policy function.
2. Calculate the reservation wage for the values $b = 10, b = 30, b = 50$. Is the reservation wage increasing or decreasing in b ?
3. Calculate the reservation wage for the values $\beta = 0.8$ and $\beta = 0.99$. Is the reservation wage increasing or decreasing in β ?
4. Suppose the worker is risk-averse with utility function $u(c) = \log(c)$. Calculate the reservation wage and compare it to the reservation wage for where the worker is risk-neutral.

Question 2 - McCall Model with infrequent offers

Consider the same environment as above, except now, the worker receives a job offer with only probability $\phi = 0.7$ and no offer with probability $1 - \phi = 0.3$.

1. Write down the Bellman equations for the worker's problem. (**Hint:** there are now two Bellman equations - one for where the unemployed worker gets a wage offer w , and one for where the unemployed worker receives no wage offer at all.)
2. Compute the reservation wage (assuming the same parameter values as in Question 1). Compare it to the case where $\phi = 1$.